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DATA STORAGE REIMAGINED

Solution Proposal for

PELOTON

US Solution V1 (400TB Optimum with HA) 07/20/2022

Your Account team

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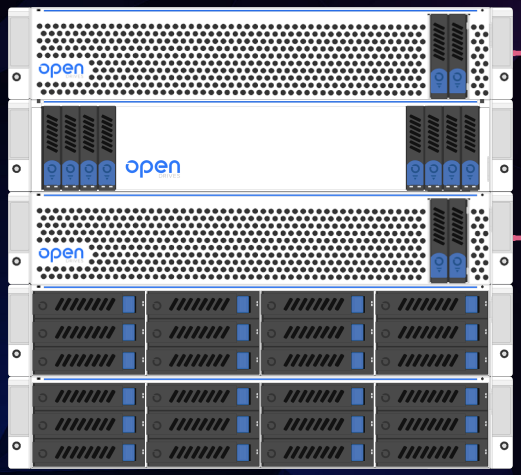
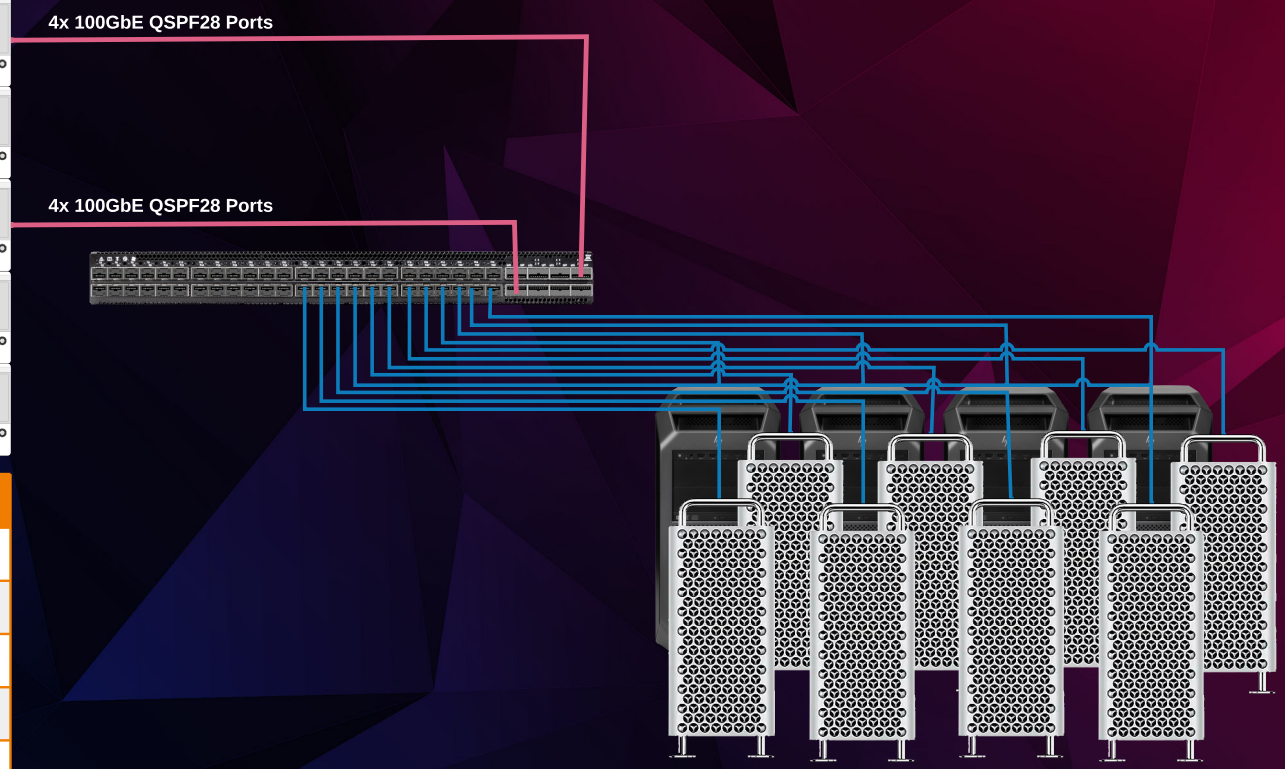
Proposed Solution



Optimum Series
2 Optimum 15 C Modules (HA)
1TB of RAM
1 F2 Module - 8 x 1.92TB NVMe Drives
7.7 uTB of NVMe Cache
5 uTB NVMe Pool
2 H16 Modules - 42 X 16TB Drives
395 uTB of HDD



NAS Connected Workstations

Requirement of ~1.3 GB/s

- 2 Ingest servers recording 8 channels each of ProRes HD 59.94 42MB/s per stream
- 9 Editors Working 1 streams each of ProRes HD 59.94 42MB/s per stream
- 1 Editors Working 5 streams each of ProRes HD 59.94 42MB/s per stream

As Configured

- Provides up to 5GB/s (expandable)
- Provides 400TB of usable capacity (expandable)
  (395TB usable of HDD and 5 usable TB of NVMe)

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ultra
HARDWARE PLATFORM

Optimum Series

When flexibility and options are key

Functional Summary

The Optimum enterprise storage solution combines architectural flexibility and scalability with the ability to select all-flash NVMe and HDD capacity modules. The Optimum is intended for businesses needing multiple storage capacity options while leveraging the performance of this workhorse storage solution. Powered by our Atlas software, the Optimum offers not only performance and scale but also provides robust data integrity, ease of use, and transparent centralized management. The following table includes the Optimum's operational specifications.

Technical Specifications

Optimum 15 overview	
Throughput	Up to 15GB/s
Network protocols	SMB up to v3.1.1
	NFS up to v4.2
	S3 object API
Notification & monitoring	SMTP, Webhooks/API, SNMP, OpenMetrics and more
Management	Secure web interface, RESTful API

Configuration				
CPU	8 cores @ 3.8 GHz			
RAM	1 TB DDR4 ECC			
	Ports	Speed	Connector	Supported Formats
Ethernet connectivity	4	100 Gbps	QSFP28	1/10/25/40/50/100 GbE
	2	10 Gbps	RJ45	1/10GbE
Out-of-band management	IPMI 2.0/Redfish			
System drives	Mirrored hot-swappable SSDs			
Supported platforms	Atlas 1.5 or later			
High availability compatible	Yes			
	F/FD	H	HD/X	
Maximum storage modules	1	16	8	

Dimensions	
Form factor	2 rack units (RU)
Physical height	3.5 in (8.9 cm)
Physical width	17.5 in (44.5 cm)
Physical depth	21 in (53.3 cm)

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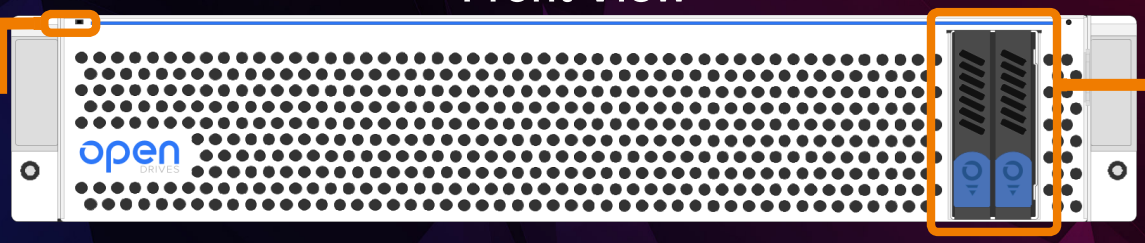
HARDWARE PLATFORM

Optimum 15
Hardware View

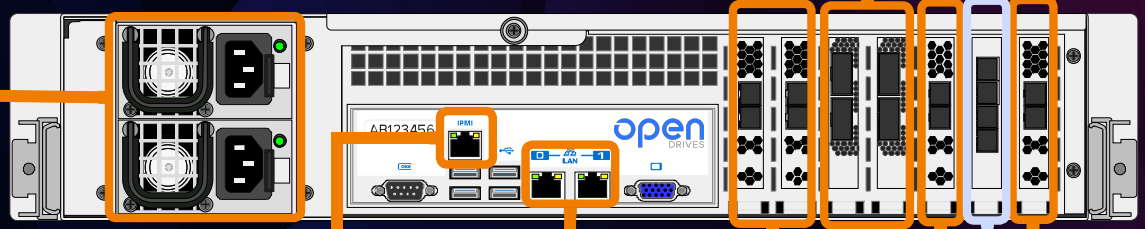
Front View



Components
Power Button
LED Status Bar
Removable Faceplate

System Drives
Mirrored Atlas SSDs

Rear View



Power Supplies
2 x 800W

Network Ports
NIC Slots: 2
Total Ports: 4
100 GbE QSFP28

IPMI Port
1 x 1 Gbps Ethernet

Network Ports
2 x 1/10 Gbps Ethernet

Expansion Ports
8 X 12Gb SAS Ports
4 X NVMe (PCIe) Ports

Other Specifications

Dimensions
Rack Units: 2U
Height: 3.5 in (88.9 mm)
Width: 17.5 in (444.5 mm)
Depth: 20.1 in (510.5 mm)

Power Usage 120V Steady (Avg)
Amps: 2.92
Watts: 350
BTU/hr: 1,194

Power Usage 240V Steady (Avg)
Amps: 1.43
Watts: 343
BTU/hr: 1,170

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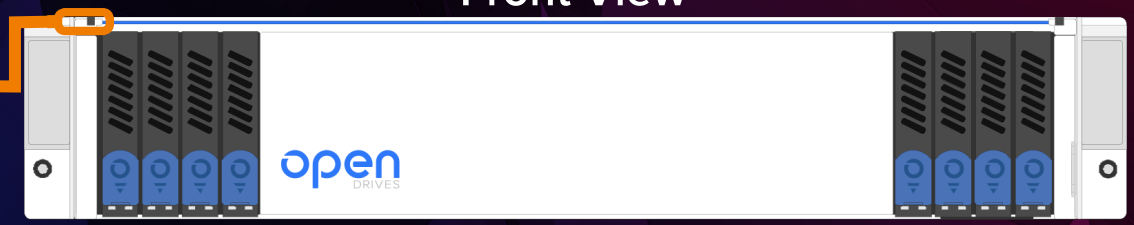
F Module

Ultra Hardware View

Front View

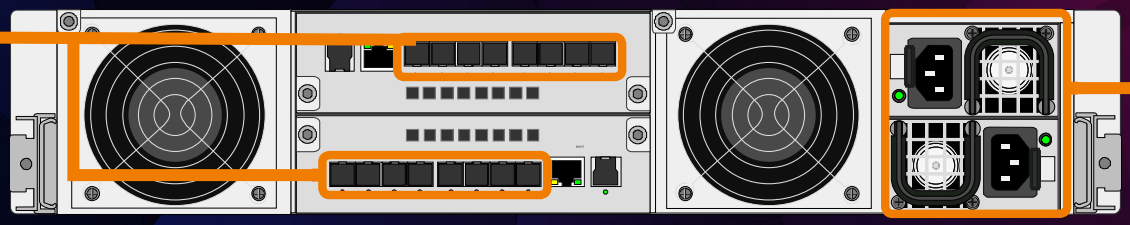
Components
Power Button
LED Status Bar
Removable Faceplate



Drives
NVMe Count: 8
NVMe Size 2.5" U.2

Rear View

Expansion Ports
16 x NVMe (PCIe) Ports



Power Supplies
2 x 800W

Options	F2	F4
Drive sizes	1.92 TB	3.84 TB
Pool usable capacity*	5 TB	10
Cache capacity	7.7 TB	15.4

*Based on 4 of the 8 drives in OpenDrives single parity pool

Other Specifications

Dimensions
Rack Units: 2U
Height: 3.5 in (88.9 mm)
Width: 17.5 in (444.5 mm)
Depth: 13.4 in (340.4 mm)

Power Usage 120V Steady (Avg)
Amps: 1.83
Watts: 220
BTU/hr: 751

Power Usage 240V Steady (Avg)
Amps: 0.90
Watts: 216
BTU/hr: 736

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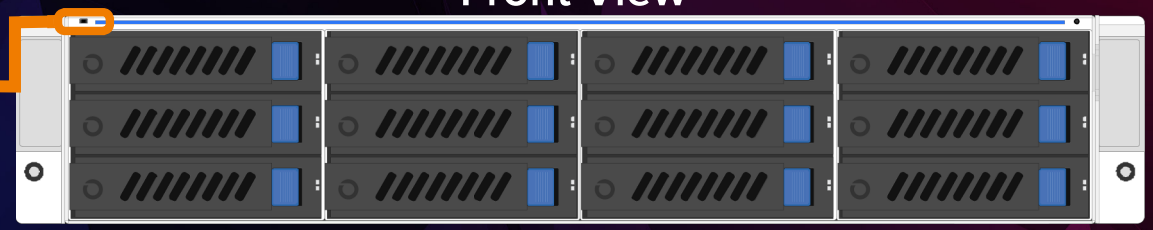
H Module

Ultra Hardware View

Front View

Components
Power Button
LED Status Bar
Removable Faceplate

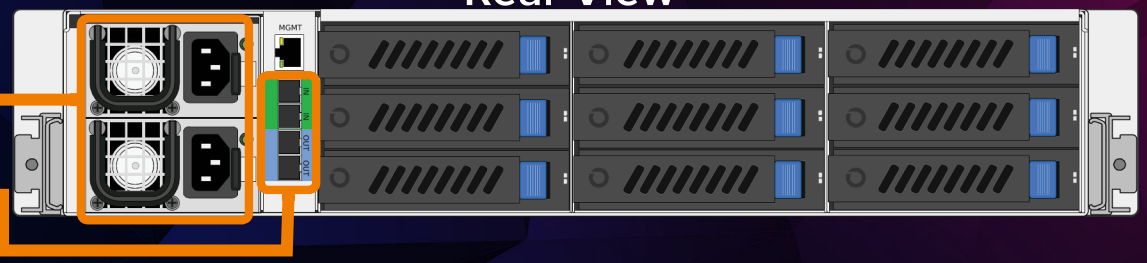


Drives
HDD Count: 12
HDD Size 3.5"

Rear View

Power Supplies
2 x 800W

Expansion Ports
4 X 12Gb SAS Ports



Drives
HDD Count: 9
HDD Size 3.5"

Options	H4	H8	H12	H16
Drive sizes	4 TB	8TB	12 TB	16 TB
Pool usable capacity*	50 TB	100 TB	150 TB	200 TB

*Based on all drives in OpenDrives standard dual parity pool

Other Specifications

Dimensions
Rack Units: 2U
Height: 3.5 in (88.9 mm)
Width: 17.5 in (444.5 mm)
Depth: 20.1 in (510.5 mm)

Power Usage 120V Steady (Avg)
Amps: 2.08
Watts: 250
BTU/hr: 853

Power Usage 240V Steady (Avg)
Amps: 1.02
Watts: 245
BTU/hr: 836

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atlas

CORE

Software that drives storage innovation

Performance Features

- Inline Caching
- Active Data Prioritization
- Dynamic Block
- Inline Compression
- Intelligent Pre-fetching

Integrity / Reliability

- Clustering
- High Availability
- Snapshots & Block Replication
- Fully Check-summed
- Encryption

Scale

- Massively Scalable
- Scale-Up
- Scale-Out
- Distributed File System

Ease of Use

- Single Namespace
- Single Pane of Glass
- Standards-Based Protocols

Containerization

- Pods
- Recipes

Analytics

- Real-time Analytics
- Flexible Notifications

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Proposed Solution

TOTAL

400TB usable in 10RU

OpenDrives Optimum Series HA

2 Optimum Compute Modules
1 F Module
2 H Modules
100GbE x 8 (MPO/OM4)*
PCIe Cables x8
SAS3 Cables x8
USB Cables x2

Ethernet Switch
H Module
Optimum 15 C Modules
F Module
Optimum 15 C Modules
H Module

*Purchase of server side optics is the sole responsibility of the customer

	Watts 120V	Amps 120V	BTU/hr 120V
Peak	1875	15.63	6398
Steady	1420	11.83	4845
Idle	960	8.00	3276

	Watts 240V	Amps 240V	BTU/hr 240V
Peak	1838	7.66	6270
Steady	1392	5.80	4748
Idle	941	3.92	3210

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Support Program

Services Summary

The OpenDrives Support Program provides hardware and software support for customers, including technical support, remote and on-site response, remote monitoring, and installation of non-customer replaceable units.

The OpenDrives Support Program includes:

- Ability to initiate support requests via multiple channels, including online chat, web, telephone, or email
- Access to Online Support Tools:
  - 24/7 Access to online support tools through OpenDrives Support Portal and knowledge base
  - Access to information from the OpenDrives Interoperability Lab
  - Connect with OpenDrives users and experts through Support Community Forums
- RMA replacement parts - Replacements parts for CRU and FRU components will be shipped to the customer location. Customers are responsible for the installation of parts deemed to be customer replaceable units (CRU's) as determined by OpenDrives.
- RMA replacement parts with an on-site response for parts deemed non-customer replaceable (FRUs) as determined by OpenDrives.

Extended Support

The OpenDrives Support Program provides hardware and software support for customers, including technical support, remote and on-site response, remote monitoring, and installation of non-customer replaceable units.

Program Features	Description	Coverage Details
Technical Support	OpenDrives provides (i) a technical response by remote means based on the Severity Level of the problem; or, (ii) when deemed necessary by OpenDrives, an onsite Response as described below.	Technical Support and initial technical response are available during standard business hours. The initial technical response is based upon Severity Level, within the following time period after receipt of Customer contact: Severity Level 1: 4 business hours Severity Level 2: 8 business hours Severity Level 3: 12 business hours
On-site Response	OpenDrives will send authorized personnel to the storage location to perform support duties after OpenDrives has isolated the problem and deemed On-Site Response necessary. NOTE: Onsite response may not be available in all geographic regions.	Initial On-site Response is based on Severity Level, within the following time period after OpenDrives deems Onsite Support is necessary. Severity Level 1: 8 business hours Severity Level 2: 1 business day Severity Level 3: 2 business days
Replacement Parts	OpenDrives provides replacement parts for products covered by a support plan when deemed necessary by OpenDrives.	Replacement part delivery objective is based upon Severity Level, within the following time period after OpenDrives deems a replacement part is necessary: Severity Level 1: 1 business day Severity Level 2: 2 business days Severity Level 3: 3-5 business days **NOTE**: Local country shipment cut-off times may impact the same day/ next local business day delivery of replacement parts and the related Onsite Response. The customer is responsible for returning the replaced CRU to a facility designated by OpenDrives unless otherwise indicated.
Installation of Software Releases	OpenDrives will perform the installation of new Software Releases as they become available for supported systems.	OpenDrives will install or assist in installing eligible and compatible OpenDrives software on the OpenDrives system. Software upgrade entitlement is determined by the Software License for the system, which is separate from the support program.
Remote Monitoring	Certain OpenDrives products will automatically and independently contact OpenDrives to provide input to assist OpenDrives engineers in problem determination. OpenDrives will remotely access products if necessary for additional diagnostics and to provide remote technical support.	For products that have remote monitoring tools and technology available from OpenDrives. Once OpenDrives is notified of a problem, the same response objectives for Technical Support and Onsite Response as previously described will apply.
Access to Online Support Tools	OpenDrives online support tools include a portal-based system for opening and managing tickets, support and workflow focused knowledge base, and a community forum where OpenDrives users and experts can communicate.	Available to all OpenDrives customers with active support agreements.

! **11/2020 update:** Due to COVID-19 OpenDrives is currently performing contactless support, and working with on-site personnel for support issues that require an on-site presence. **Please contact us for details.**

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